

Nurture Course

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL 2023



Test Type : **PRACTICE TEST**

Test Pattern : **NEET (UG)**

TEST DATE :

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	3	4	1	2	2	2	4	2	4	1	1	2	4	2	2	3	2	3	1	1	2	3	1	1	2	3	2	4	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	2	4	4	1	3	3	1	1	3	1	1	1	2	3	1															

HINT - SHEET

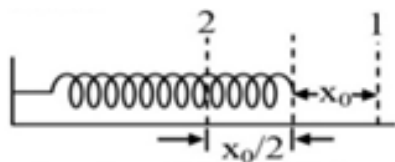
1. **Ans (1)**

$$F = ma = \frac{mv}{T} \quad (a = \frac{v-0}{T})$$

Instantaneous power

$$Fv = mav = \frac{mv}{T}at = \frac{mv}{T} \frac{v}{T}t = \frac{mv^2}{T^2}t$$

2. **Ans (3)**



Change in potential energy is independent of reference

$$U_2 - U_1 = \frac{1}{2}k \left(\frac{x_0}{2} \right)^2 - \frac{1}{2}kx_0^2 = -\frac{3}{8}kx_0^2$$

3. **Ans (4)**

From work energy theorems,

$$\frac{p^2}{2m} = \mu mgs \Rightarrow S = \frac{p^2}{2\mu m^2 g}$$

4. **Ans (1)**

Gain in KE = Loss in PE

$$\frac{1}{2}mv^2 = U \Rightarrow m = \frac{2U}{V^2}$$

5. **Ans (2)**

At the bottom most point, square of speed of bob

$$v^2 = 2gL(1 - \cos \alpha)$$

It will rise further to a height

$$h = \frac{v^2}{2g} = L(1 - \cos \alpha)$$

$$\text{or } (L - l)(1 - \cos \theta) = L(1 - \cos \alpha)$$

$$\therefore \theta = \cos^{-1} \left[\frac{L \cos \alpha - l}{L - l} \right]$$

6. **Ans (2)**

$$P = Fv = C \text{ (constant)}$$

$$Fv = C$$

$$maV = C$$

$$v \frac{dv}{dt} = \frac{C}{m}$$

$$\int v dv = \frac{C}{m} \int dt$$

$$\frac{v^2}{2} = \frac{C}{m}t$$

$$V = \left(\frac{2C}{m} \right)^{1/2} t^{1/2}$$

$$\left(\frac{2C}{m} \right)^{1/2} = \text{constant} = k$$

$$v = kt^{1/2}$$

$$\frac{ds}{dt} = kt^{1/2}$$

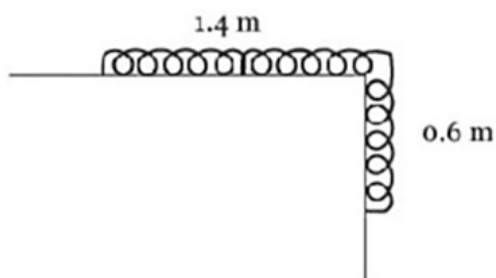
$$\int ds = k \int t^{1/2} dt$$

$$S \propto t^{3/2}$$

7. **Ans (2)**

In case of non-conservative forces, the work done is dissipated as heat, sound, etc. i.e., it does not increase the potential energy. But in case of conservative forces, work done is responsible for increasing or decreasing the potential energy.

8. **Ans (4)**



The center of mass of the hanging part is at 0.3 m from table

Mass of hanging part = $4 \times 0.6 = 1.2$ kg

So, $W = mgh = 1.2 \times 10 \times 0.3 = 3.6$ J

(center of hanged part distance from top is taken as several points are located at different distance from top)

9. **Ans (2)**

Since $K_1 > K_2$, therefore, P will develop greater restoring force than Q. So the applied force on P will be more than the applied force on Q. Extension is same in both the cases. So, more work is done on P.

10. **Ans (4)**

$$x = \frac{t^4}{4} + 3$$

$$\frac{dx}{dt} = \frac{4t^3}{4} + 0$$

$$V = t^3$$

Using work energy theorem

$$w = \Delta k = \frac{1}{2}m(v(2))^2 - \frac{1}{2}m(v(0))^2$$

$$= \frac{1}{2}(2) [(2^3) - 0] = 64\text{ J}$$

11. **Ans (1)**

$$U(x) = (x^2 - 3x)\text{ J}$$

For a conservative field, Force $F = -\frac{du}{dx}$

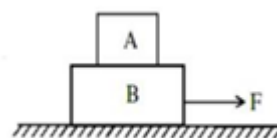
$$\therefore F = -\frac{d}{dx}(x^2 - 3x) = -(2x - 3) = -2x + 3$$

At equilibrium position, $F = 0$

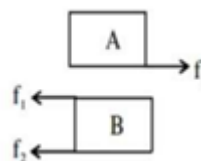
$$\Rightarrow -2x + 3 = 0 \Rightarrow x = \frac{3}{2}\text{ m} = 1.5\text{ m}$$

12. **Ans (1)**

Consider the blocks shown in the figure to be move together due to static friction between them. The free body diagram of both the blocks are shown below.



Work done by static friction on A is positive and on B is negative



13. **Ans (2)**

Mass is displaced from (0,0) to (1,1). Therefore displacement is,

$$\vec{S} = (1 - 0)\hat{i} + (1 - 0)\hat{j}$$

$$\vec{S} = (\hat{i} + \hat{j})\text{ m}$$

Given force

$$\vec{F} = (4\hat{i} + 4\hat{j})\text{ N}$$

Work done by a constant force,

$$W = \vec{F} \cdot \vec{S}$$

$$W = (4\hat{i} + 4\hat{j}) \cdot (\hat{i} + \hat{j})$$

$$W = 4 + 4 = 8\text{ J}$$

From work energy theorem - $\Delta K = W$

$$\frac{1}{2}m(v_2^2 - v_1^2) = W$$

$$\frac{1}{2}(1)(v_2^2 - 2^2) = 8$$

$$v_2^2 = 20$$

$$v_2 = 4.5\text{ m/s}$$

14. **Ans (4)**

$$\text{Power} = \frac{\text{work}}{\text{time}} = \frac{\text{energy}}{\text{time}} = \frac{mgh}{t} = \frac{Wh}{t}$$

(W = weight)

$$\therefore \frac{P_1}{P_2} = \frac{W_1}{W_2} \times \frac{t_2}{t_1} = \frac{4}{3} \times \frac{11}{12} = \frac{11}{9}$$

(h is same in both cases)

15. **Ans (2)**

$$F = cx$$

$$W = \int^F dx$$

$$W = \int_0^{x_1} cxdx$$

$$F = \left[c \frac{x^2}{2} \right]_0^{x_1}$$

$$F = c \frac{x_1^2}{2}$$

16. **Ans (2)**

According to Work Energy Theorem work done by all forces is equal to change in kinetic energy.

17. **Ans (3)**

We know that $U = ax + by$

$$\frac{\partial U}{\partial x} = \frac{\partial}{\partial x}(ax + by)$$

$$\frac{\partial U}{\partial x} = a$$

$$\frac{\partial U}{\partial y} = \frac{\partial}{\partial y}(ax + by)$$

$$\frac{\partial U}{\partial y} = b$$

$$\vec{F} = -\left(\frac{\partial U}{\partial x} \hat{i} + \frac{\partial U}{\partial y} \hat{j}\right)$$

$$\vec{F} = -a\hat{i} - b\hat{j}$$

$$a = \frac{F}{m} = \frac{\sqrt{a^2 + b^2}}{m}$$

18. **Ans (2)**

Work done by both the persons is the same and is given as, $W = mgh$, It does not depend upon the time taken to perform the work.

19. **Ans (3)**

Let work done by all forces other than conservative forces = W_{others}

Work done by conservative forces, W_{con} is negative of change in potential energy.

$W_{\text{con}} = -\Delta U$, From work energy theorem,

$$W_{\text{by all forces}} = \Delta K$$

$$W_{\text{others}} + W_{\text{con}} = \Delta K$$

$$W_{\text{others}} - \Delta U = \Delta K$$

$$W_{\text{others}} = \Delta U + \Delta K$$

Therefore, work done by all forces other than conservative forces is equal to change in total mechanical energy.

20. **Ans (1)**

From work energy theorem

$$W = \Delta K$$

$$\int_0^2 Fdx = \frac{1}{2}m(v^2 - u^2)$$

$$\int_0^2 (3x^2 - 12x)dx = \frac{1}{2}2(v^2 - (4\sqrt{5})^2)$$

$$[x^3 - 6x^2]_0^2 = v^2 - 80$$

$$8 - 24 + 80 = v^2$$

$$v = 8\text{ms}^{-1}$$

21. Ans (1)

Mass of the body (m) = 0.5 kg

Velocity of the body (v) = $ax^{\frac{3}{2}}$

Where a = 5

Velocity of the body at x=0,

$$V_1 = 5 \times 0 = 0$$

Velocity of the body at x=2,

$$V_2 = 5 \times \left(2^{\frac{3}{2}}\right) \text{ m/s}$$

According to work energy theorem,

Work done = change in K.E.

$$= \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

$$= \frac{1}{2}m(v_2^2 - v_1^2) = \frac{1}{2} \times 0.5 \left[\left(5 \times 2^{\frac{3}{2}}\right)^2 - 0^2 \right]$$

$$= \frac{1}{2} \times 0.5 \times 25 \times 8 = 50\text{J}$$

22. Ans (2)

In the potential energy curve, the point at maximum (crest) is the point of unstable equilibrium since, the displacement of particle towards the left or right will experience the increase or decrease in the force and potential energy, both are related by the relation,

$$F = -\frac{\partial U}{\partial x}, \text{ Hence, the point at unstable}$$

equilibrium is point B

23. Ans (3)

The spring will exert maximum force when the ball is at its lowest position. If the ball has descended through a distance x to reach the position.

$$mgx = \frac{1}{2}kx \text{ or } x = \frac{2mg}{k} \dots\dots (i)$$

For block B to leave the contact spring force

$$kx = Mg \dots\dots(ii)$$

Comparing both the equation m = M/2

24. Ans (1)

$$\frac{dv}{dt} \propto -x \Rightarrow \frac{dv}{dx} \cdot \frac{dx}{dt} = -cx$$

$$Vdv = -cx dx$$

$$\frac{v^2}{2} = -\frac{cx^2}{2} \text{ kinetic energy} = \frac{mv^2}{2}$$

$$\text{kinetic energy} = \frac{mcx^2}{2}$$

$$k \propto x^2$$

25. Ans (1)

Nail has fallen through a total vertical distance of (h+x) hence loss in its

Potential energy = mg(h+x). If average retarding force exerted by sand on the nail is

F, then work done - W = -Fx

For equilibrium, mg(h+x) = Fx

$$F = mg \frac{h+x}{x} = mg \left(\frac{h}{x} + 1 \right)$$

26. Ans (2)

$$W = \int_{(0,0)}^{(1,1)} \vec{F} \cdot d\vec{x}$$

$$\text{Here } d\vec{s} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$W = \int_{(0,0)}^{(1,1)} (x^2 dy + y dx) = \int_{(0,0)}^{(1,1)} (x^2 dy + x dx)$$

as (x=y)

$$W = \left[\frac{y^3}{3} + \frac{x^2}{2} \right]_{(0,0)}^{(1,1)} = \frac{5}{6} \text{ J}$$

27. Ans (3)

Loss in P.E. = gain in K.E. + work done against friction

$$mgR = \frac{1}{2}m(1.4gR) + W_f$$

$$W_f = 0.3mgR$$

$$\text{Now } W_{B \rightarrow C} = mgR + 0.3mgR = 1.3mgR$$

28. **Ans (2)**

Only external force can change the momentum of the system while internal forces can change the momentum of individual particles and hence the kinetic energy of the system.

29. **Ans (4)**

For the upper half

$$v^2 = u^2 + 2al/2 = 2(g\sin\theta)l/2 = gl\sin\theta \dots(1)$$

For the lower half

$$\Rightarrow 0 = u^2 + 2g(\sin\theta - \mu\cos\theta)\frac{1}{2}$$

$$a = v^2 + 2g(\sin\theta - \mu\cos\theta)\frac{1}{2} \dots\dots(2)$$

From (1) & (2)

$$\Rightarrow -gl\sin\theta = gl(\sin\theta - \mu\cos\theta)$$

$$\Rightarrow \mu\cos\theta = 2\sin\theta \Rightarrow \mu = 2\tan\theta$$

30. **Ans (2)**

Loss of energy = 35%

Remaining energy = 65%

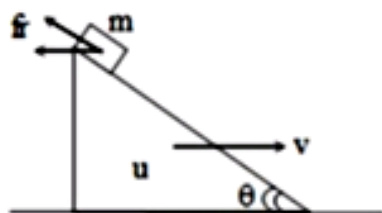
$\Rightarrow$ New energy = 65% of initial energy

$$Mgh = \frac{65}{100}$$

Mgh'

$$h' = \frac{65}{100} \times 100 = 65m$$

31. **Ans (2)**



As the block is rest with respect to the wedge,

$$f_r = mg \sin \theta$$

Work done by friction

$$W = FScos\theta$$

$$W = -mg \sin \theta (vt) \cos \theta$$

$$W = -\frac{mg \sin 2\theta}{2} vt$$

32. **Ans (4)**

$$W = - (Mg \sin \theta) d$$

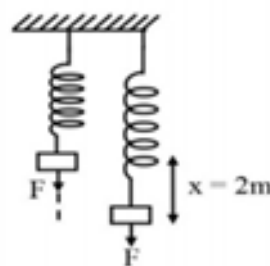
33.

Ans (4)

K.E is a scalar. Since it is zero, every individual velocity is zero. Momentum is a vector. If it is zero, it does not imply every velocity of particles in the system is equal to zero

34.

Ans (1)



At equilibrium, $mg = kx_0$

$$\Rightarrow x_0 = \frac{mg}{k} = 1m$$

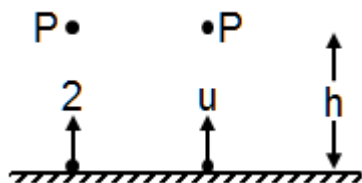
$$\therefore W_{\text{ext}} = U_2 - U_1 (W_{\text{ext}} = U_2 - U_1)$$

$$= \frac{1}{2}kx_2^2 - \left[\frac{1}{2}kx_1^2 + mgh \right]$$

$$= \frac{1}{2}k(x_2^2 - x_1^2) - mgh$$

$$= \frac{1}{2} \times 100 \times (3^2 - 1^2) - 10 \times 10 \times 2 = 200J$$

35. Ans (3)



By energy conservation between (A & P)

$$k + 0 = mgh + k$$

$$\therefore mgh = k' \text{ given}$$

$$\text{So } K = 2mgh$$

when thrown by $2u$ speed

$$K_1 = \frac{1}{2}m(2u)^2 = 4k$$

$$\text{or } K_1 = 4(2mgh) = 8mgh$$

Now energy conservation between (B & P)

$$K_1 + 0 = mgh + K.E.$$

$$8mgh - mgh = KE$$

$$\text{So, } \frac{PE}{KE} = \frac{mgh}{7mgh} = \frac{1}{7}$$

36. Ans (3)

$$F = 3x + 5$$

$$W = \int_2^6 F \cdot dx = \int_2^6 (3x + 5) dx$$

$$\begin{aligned} &= \left[\frac{3x^2}{2} + 5x \right]_2^6 = \left[\frac{3}{2} \times [36] + 30 \right] - \left[\frac{3}{2} \times 4 + 10 \right] \\ &= 84 - 16 \Rightarrow 68 \text{ J} \end{aligned}$$

37. Ans (1)

$$u = k(x + y)$$

$$W_{\text{cons}} = -\Delta u$$

$$= -[u_f - u_i] = -[k(5) - 2k] = -3k$$

38. Ans (1)

$\frac{3}{4}$ th energy is lost i.e. $\frac{1}{4}$ th kinetic energy is left. Hence, its velocity becomes

$$\frac{v_0}{2} \text{ under a retardation of } \mu g \text{ in time } t_0$$

$\therefore$ from the first equation of motion

$$\frac{v_0}{2} = v_0 - \mu g t_0$$

$$\text{Or } \mu g t_0 = \frac{v_0}{2}$$

$$\text{Or } \mu = v_0 / 2g t_0$$

39. Ans (3)

In a conservative field $F = -\frac{dU}{dr}$

$$F = -\frac{d}{dx}(ax^2 - bx) = b - 2ax$$

40. Ans (1)

$$P = mg \sin \theta \times V = 4000W$$

41. Ans (1)

From work energy theorem for upward motion

$$\frac{1}{2}m(16)^2 = mgh + W$$

For downward motion,

$$\frac{1}{2}m(8)^2 = mgh - W \Rightarrow h = 8m$$

42. Ans (1)

Energy stored in a spring is given by

$$U = \frac{1}{2}kx^2$$

Where k is spring constant, Now applying

Hooke's law, we have

$$\text{Tension } T = kx \Rightarrow \text{Extension } x = T/k \Rightarrow$$

$$U = \frac{T^2}{2k}$$

43. Ans (2)

$$v = \frac{dx}{dt} = \frac{1}{4} \times 4t^3 = t^3 = 1 \text{ ms}^{-1}$$

work done = change in K.E.

$$= \frac{1}{2} \times 1 \times 1 - \frac{1}{2} \times 1 \times 0 = \frac{1}{2} \text{ J}$$

44. Ans (3)

$$P = Fv \quad \text{Or} \quad P = \left(mv \frac{dv}{dx} \right) v$$

$$\text{or} \int_0^v v^2 dv = \int_0^x \frac{P}{m} dx$$

$$\frac{v^3}{3} = \frac{Px}{m} \quad \text{or} \quad v = \left(\frac{3xP}{m} \right)^{1/3}$$

45. Ans (1)

Weight of train $10^7 \text{ N} = 10^6 \text{ kgf}$

Frictional force, $F = \frac{0.5}{100} \times 10^6 = 5000 \text{ kgf}$

$F = 50000 \text{ N}$

$v = 36 \text{ km/h} = 10 \text{ m/s}$

$P = F \times v = 50000 \times 10$

$= 5 \times 10^5 \text{ W} = 500 \text{ kW}$